LECTURE 08

Natural Language Processing

Model Evaluation Metrics
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PRECISION • RECALL • F1-SCORE

Today’s Objective Is Measuring What the Model Gets Right

From predictions to meaningful evaluation

STEP 01

PREDICT

Generate labels or probabilities.

STEP 02

COUNT

Compare predictions with ground truth.

STEP 03

MEASURE

Compute accuracy, precision, recall and F1.

STEP 04

INTERPRET

Decide which errors matter for the application.

THE GOAL

Move beyond “accuracy” and understand what kinds of mistakes the model makes.

Evaluation Completes the NLP Learning Pipeline

From representation and learning to evidence about generalization

LECTURE 03	**Text Representation**	Text → BoW / TF-IDF
LECTURE 04	**Naïve Bayes**	Probabilistic classification
LECTURE 05	**Perceptron**	Linear discriminative learning
LECTURE 06	**Logistic Regression**	Probabilities + Log Loss
LECTURE 07	**Regularization & Optimization**	Control complexity + learn parameters
TODAY	**Model Evaluation**	Measure performance and error types

EVALUATION IS THE FEEDBACK LOOP

A Prediction Is Not Enough

A classifier produces an answer; evaluation tells us whether that answer is useful.

MODEL OUTPUT

Positive / Negative / Class A / Class B

The prediction alone does not tell us whether it is correct.

GROUND TRUTH

The trusted label assigned by the dataset, annotator, or task definition.

EVALUATION

Compare predictions against ground truth and quantify the errors.

The Confusion Matrix Is the Foundation

PREDICTED

ACTUAL		
	TRUE POSITIVE **TP** Correct Positive	**FALSE NEGATIVE** **FN** Missed Positive
	FALSE POSITIVE **FP** False Alarm	**TRUE NEGATIVE** **TN** Correct Negative

Rows = actual class • Columns = predicted class

Four Outcomes Capture Every Binary Prediction

ACTUAL

PREDICTED

TP Actual + Predicted +	**FN** Actual + Predicted −
FP Actual − Predicted +	**TN** Actual − Predicted −

KEY IDEA

Every binary prediction belongs to exactly one of these four categories.

True Positive Means the Model Correctly Finds th

TP

Actual Positive + Predicted Positive

EXAMPLE

A spam email is correctly classified as spam.

WHY IT MATTERS

True positives are the successes we want to increase.

True Negative Means the Model Correctly Rejects

TN

Actual Negative + Predicted Negative

EXAMPLE

A legitimate email is correctly classified as not spam.

WHY IT MATTERS

True negatives are important when false alarms are costly.

False Positive Means a False Alarm

FP

Actual Negative + Predicted Positive

EXAMPLE

A legitimate email is incorrectly flagged as spam.

WHY IT MATTERS

Too many false positives make a system over-alert and inconvenient.

False Negative Means a Missed Target

FN

Actual Positive + Predicted Negative

EXAMPLE

A spam email passes through the inbox.

WHY IT MATTERS

False negatives are missed targets; they can be costly in retrieval or detection tasks.

Accuracy Measures Overall Correctness

ACCURACY

How many predictions are correct overall

FORMULA

Accuracy = (TP + TN) / (TP + TN + FP + FN)

STRENGTH

Easy to understand and useful when classes are reasonably balanced.

LIMITATION

Can hide poor performance on a minority class.

Accuracy Is Simple—But Can Be Misleading

CONFUSION MATRIX

TP = 85
TN = 90
FP = 15
FN = 10

CALCULATION

Accuracy = (85 + 90) / 200
= 0.875
= 87.5%

CAUTION

87.5% sounds strong, but we still need to know whether the errors are concentrated in one class.

Precision Asks: When the Model Says Positive, Is It Right?

PRECISION

Of everything predicted Positive, how much was actually Positive?

FORMULA

Precision = TP / (TP + FP)

INTERPRETATION

High precision means the model's Positive predictions are usually trustworthy.

Precision Focuses on the Quality of Positive Predictions

HIGH PRECISION

TP = 90
FP = 10
Precision = 90 / 100 = 90%

Only 1 in 10 positive alerts is a false alarm.

LOW PRECISION

TP = 50
FP = 50
Precision = 50 / 100 = 50%

Half of the positive alerts are wrong.

USEFUL WHEN

False positives are especially costly.

Recall Asks: Did the Model Find the Positives?

RECALL

Of all actual Positives, how many did the model find?

FORMULA

Recall = TP / (TP + FN)

INTERPRETATION

High recall means the model misses relatively few examples of the Positive class.

Recall Focuses on Missing as Few Positives as Possible

HIGH RECALL

TP = 95
FN = 5
Recall = 95 / 100 = 95%

The model finds almost all positives.

LOW RECALL

TP = 50
FN = 50
Recall = 50 / 100 = 50%

Half of the actual positives are missed.

USEFUL WHEN

False negatives are especially costly.

Precision and Recall Often Pull in Opposite Directions

PUSH FOR RECALL

Lower the decision threshold.

More cases become Positive → fewer misses, but more false alarms.

PUSH FOR PRECISION

Raise the decision threshold.

Fewer cases become Positive → fewer false alarms, but more misses.

There is no universally “best” balance.

F1-Score Balances Precision and Recall

F1-SCORE

A single score that rewards both precision

FORMULA

F1 = 2 × Precision × Recall
/ (Precision + Recall)

KEY PROPERTY

F1 becomes low when either precision or recall is low.

F1 Uses the Harmonic Mean

GIVEN

Precision = 0.85
Recall = 0.895

CALCULATE

F1 = 2 × 0.85 × 0.895
/ (0.85 + 0.895)
≈ 0.872

INTERPRET

F1 ≈ 87.2% summarizes the balance between precision and recall.

Class Imbalance Can Make Accuracy Look Excellent

DATASET

990 Negative
10 Positive

Always predicting Negative gives 99% accuracy.

BUT…

Positive recall = 0%

The model completely fails to detect the minority class.

LESSON

For imbalanced data, inspect precision, recall, F1 and the confusion matrix—not accuracy alone.

The Right Metric Depends on the Cost of Errors

PRECISION

Use when false positives are costly.

Example:
spam filtering where legitimate mail must

RECALL

Use when false negatives are costly.

Example:
retrieving as many relevant documents as

F1

Use when both error types matter and a single balanced score is useful.

Metric selection is a task-design decision, not merely a mathematical decision.

The Same Ideas Extend to Multiclass NLP

SPORTS

TP for Sports
FP from other categories
FN = missed Sports articles

BUSINESS

TP for Business
FP from other categories
FN = missed Business articles

POLITICS

TP for Politics
FP from other categories
FN = missed Politics articles

ONE-VS-REST VIEW

For each class, calculate its own TP, FP, FN and then aggregate the results.

Macro, Micro and Weighted Averages Answer Different Questions

MACRO

Calculate the metric separately for each c

Every class gets equal weight.

MICRO

Pool TP, FP and FN across classes first, t

Every individual example gets equal weigl

WEIGHTED

Calculate each class score, then weight by class support.

Large classes contribute more.

Macro can expose minority-class performance; weighted reflects the class distribution.

A Probability Becomes a Class Through a Decision Threshold

LOGISTIC REGRESSION OUTPUT

$$P(y = \text{Positive} \mid x) = 0.73$$

If threshold = 0.50

DECISION

$0.73 \geq 0.50$
$\longrightarrow$ Predict Positive

IMPORTANT

The threshold is a decision rule applied after the model produces a score/probability.

Changing the Threshold Changes Precision and Recall

THRESHOLD = 0.30

More examples become Positive.

Recall tends to increase.
Precision may decrease.

THRESHOLD = 0.70

Fewer examples become Positive.

Precision tends to increase.
Recall may decrease.

Threshold tuning should be based on validation data and the cost of errors.

Scikit-Learn Can Compute the Metrics Directly

PYTHON

```
from sklearn.metrics import (
    accuracy_score, precision_score,
    recall_score, f1_score
)

print(accuracy_score(y_test, y_pred))
print(precision_score(y_test, y_pred))
print(recall_score(y_test, y_pred))
print(f1_score(y_test, y_pred))
```


For multiclass problems, specify an appropriate average such as macro, micro or weighted.

A Classification Report Summarizes the Model

CLASSIFICATION REPORT

```
            precision   recall  f1-score   support

Negative        0.91      0.90      0.90       100
Positive        0.89      0.90      0.89       100

accuracy                            0.90       200
macro avg       0.90      0.90      0.90       200
weighted avg    0.90      0.90      0.90       200
```

READ IT

Per-class metrics reveal whether one class is being neglected.

Evaluation Should Lead to Error Analysis

STEP 01 — FIND ERRORS

Filter the test set to examples where predicted label ≠ true la

STEP 02 — EXPLAIN ERRORS

Look for negation, ambiguity, rare words, sarcasm, noisy labels or domain shift.

STEP 03 — IMPROVE

Change data, features, preprocessing, threshold or model only when the error pattern justifies it.

A Good NLP Evaluation Workflow Is Reproducible

01	**SPLIT**	Train / validation / test
02	**TRAIN**	Fit the model only on training data
03	**TUNE**	Use validation data for model and threshold choices
04	**TEST**	Evaluate on unseen test data
05	**ANALYZE**	Inspect metrics + confusion matrix + errors

RULE
Do not tune repeatedly on the test set; it should remain an unbiased final check.

Evaluation Turns Predictions Into Evidence

CONFUSION MATRIX

TP, TN, FP and FN describe the four possible outcomes of bina

CORE METRICS

Accuracy = overall correctness
Precision = quality of positive predictions
Recall = coverage of actual positives
F1 = balance of precision and recall

FINAL PRINCIPLE

A high score is not enough. Understand which errors occur, why they occur, and whether those errors matter.

NEXT LECTURE N-Grams & Language Modeling